CAS SciFinder

Task History

Initiating Search

November 11, 2025, 2:28 PM

Search:

CAS Lexicon Search:

Benzylic oxidation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (77)	**References**	View Results
Exported: Retrieved Related Reaction Results (25)	**Reactions**	View Results

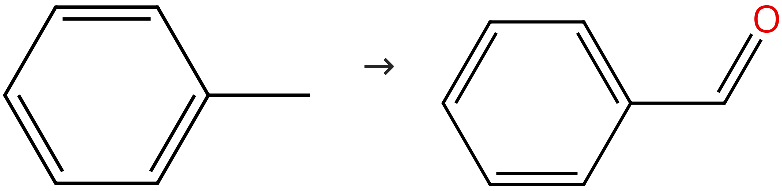
CAS SciFinder

Reactions (25)

View in CAS SciFinder

Scheme 1 (4 Reactions)

Steps: **1** Yield: **68-89%**



Suppliers (280)

Suppliers (69)

31-495-CAS-3751951

Steps: **1** Yield: **89%**

1.1 **Reagents:** Sodium hypochlorite
Catalysts: Cobalt diacetate, Tempo
Solvents: Dichloromethane; 6 h, pH 8.3, 0 - 5 °C

Experimental Protocols

Direct benzylic oxidation with sodium hypochlorite using a new efficient catalytic system: TEMPO/Co(OAc)$_2$

By: Jin, Can; et al

Synlett (2011), (10), 1435-1438.

31-495-CAS-2609459

Steps: **1** Yield: **68%**

1.1 **Reagents:** *tert*-Butyl hydroperoxide
Catalysts: Cobalt(II) acetylacetonate
Solvents: Acetone; 28 °C; 15 h, 28 °C

Experimental Protocols

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

31-614-CAS-41045910

Steps: **1**

1.1 **Reagents:** Sodium bromide, Oxygen
Catalysts: 3051562-15-0 (reaction products with ammonium acetate in oxygen)
Solvents: Acetonitrile, Water; 5 h

Experimental Protocols

Constrution of Quinazoline-Linked Covalent Organic Frameworks via a Multicomponent Reaction for Photoca talysis

By: Yu, Song-Chen; et al

Chemistry - A European Journal (2024), 30(42), e202400668.

31-495-CAS-265696

Steps: **1**

1.1 **Reagents:** Hydrogen peroxide
Catalysts: Zirconium phosphate, Cr^{3+}
Solvents: Methanol, Water; 12 h, 60 °C

Experimental Protocols

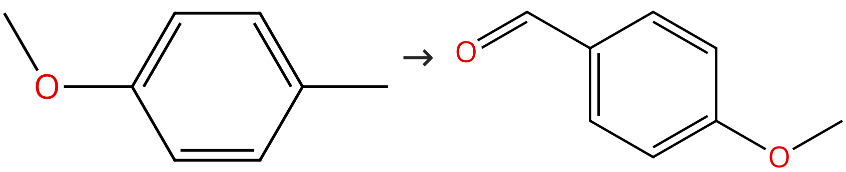
Allylic and Benzylic Oxidation over Cr ^{III}-Incorporated Mesoporous Zirconium Phosphate with 100 % Selectivity

By: Sinhamahapatra, Apurba; et al

ChemCatChem (2011), 3(9), 1447-1450.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **84%**



Suppliers (79)

Suppliers (128)

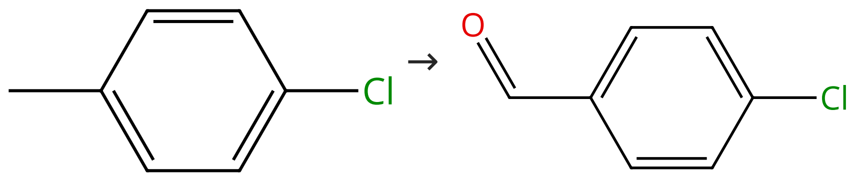
31-495-CAS-8020073	Steps: **1** Yield: **84%**
1.1	**Reagents:** Sodium hypochlorite **Catalysts:** Cobalt diacetate, Tempo **Solvents:** Dichloromethane; 6 h, pH 8.3, 0 - 5 °C
Experimental Protocols	

Direct benzylic oxidation with sodium hypochlorite using a new efficient catalytic system: TEMPO/Co(OAc)$_2$

By: Jin, Can; et al

Synlett (2011), (10), 1435-1438.

Scheme 3 (2 Reactions) — Steps: **1** Yield: **71-83%**



Suppliers (82)　　Suppliers (97)

31-495-CAS-5891406	Steps: **1** Yield: **83%**
1.1	**Reagents:** Sodium hypochlorite **Catalysts:** Cobalt diacetate, Tempo **Solvents:** Dichloromethane; 6 h, pH 8.3, 0 - 5 °C
Experimental Protocols	

Direct benzylic oxidation with sodium hypochlorite using a new efficient catalytic system: TEMPO/Co(OAc)$_2$

By: Jin, Can; et al

Synlett (2011), (10), 1435-1438.

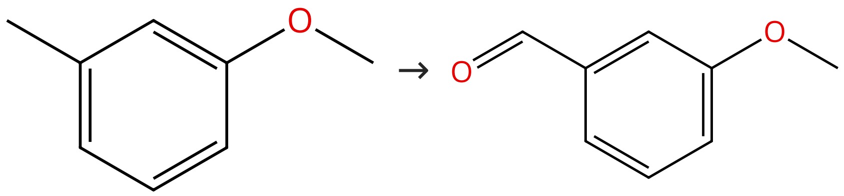
31-495-CAS-15395895	Steps: **1** Yield: **71%**
1.1	**Reagents:** *tert*-Butyl hydroperoxide **Catalysts:** Cobalt(II) acetylacetonate **Solvents:** Acetone; 28 °C; 15 h, 28 °C
Experimental Protocols	

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 4 (1 Reaction) — Steps: **1** Yield: **80%**



Suppliers (68)　　Suppliers (88)

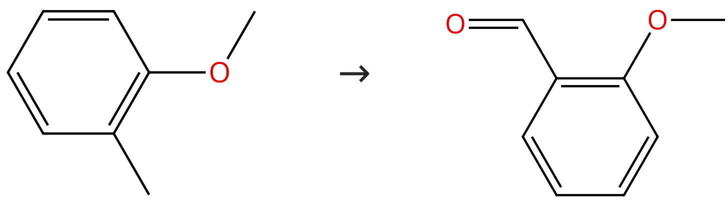
31-495-CAS-13266515	Steps: **1** Yield: **80%**
1.1	**Reagents:** *tert*-Butyl hydroperoxide **Catalysts:** Cobalt(II) acetylacetonate **Solvents:** Acetone; 28 °C; 15 h, 28 °C
Experimental Protocols	

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 5 (1 Reaction) — Steps: **1** Yield: **71%**



Suppliers (86)　　Suppliers (96)

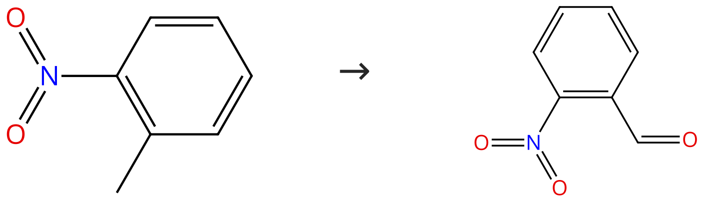
31-495-CAS-6874899	Steps: **1** Yield: **71%**
1.1	**Reagents:** *tert*-Butyl hydroperoxide **Catalysts:** Cobalt(II) acetylacetonate **Solvents:** Acetone; 28 °C; 15 h, 28 °C
Experimental Protocols	

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 6 (1 Reaction) Steps: **1** Yield: **66%**



Suppliers (61) Suppliers (99)

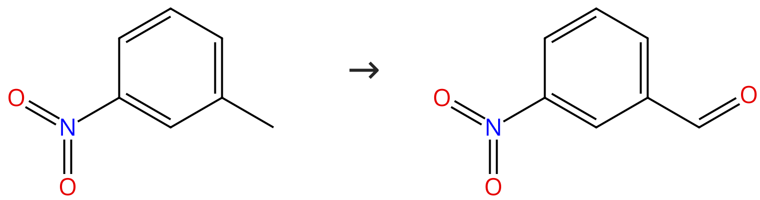
31-495-CAS-9002659	Steps: **1** Yield: **66%**
1.1	**Reagents:** *tert*-Butyl hydroperoxide **Catalysts:** Cobalt(II) acetylacetonate **Solvents:** Acetone; 28 °C; 15 h, 28 °C
Experimental Protocols	

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 7 (1 Reaction) Steps: **1** Yield: **66%**



Suppliers (54) Suppliers (89)

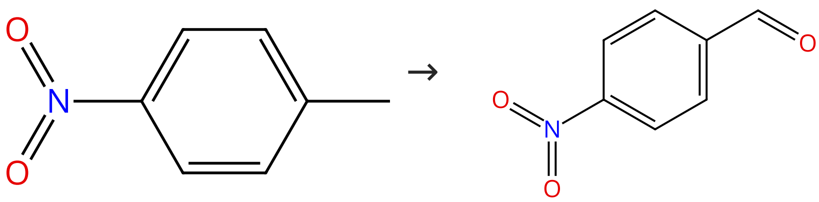
31-495-CAS-11137549	Steps: **1** Yield: **66%**
1.1	**Reagents:** *tert*-Butyl hydroperoxide **Catalysts:** Cobalt(II) acetylacetonate **Solvents:** Acetone; 28 °C; 15 h, 28 °C
Experimental Protocols	

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 8 (1 Reaction) Steps: **1** Yield: **65%**



Suppliers (75) Suppliers (91)

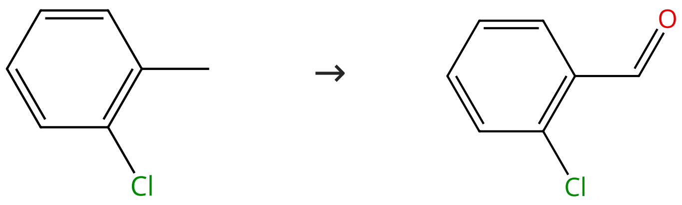
31-495-CAS-1996754	Steps: **1** Yield: **65%**
1.1	**Reagents:** *tert*-Butyl hydroperoxide **Catalysts:** Cobalt(II) acetylacetonate **Solvents:** Acetone; 28 °C; 15 h, 28 °C
Experimental Protocols	

Co(acac)$_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 9 (1 Reaction) Steps: **1** Yield: **64%**



Suppliers (71)

Suppliers (83)

31-495-CAS-4752736 Steps: **1** Yield: **64%**

1.1 **Reagents:** *tert*-Butyl hydroperoxide
Catalysts: Cobalt(II) acetylacetonate
Solvents: Acetone; 28 °C; 15 h, 28 °C

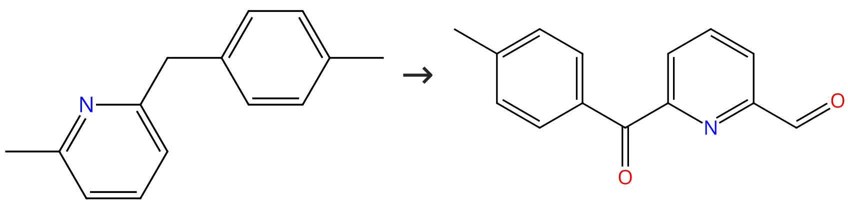
Experimental Protocols

$Co(acac)_2$-catalyzed allylic and benzylic oxidation by tert- butyl hydroperoxide

By: Han, Xiaoqiang; et al

Synthesis (2013), 45(5), 615-620.

Scheme 10 (1 Reaction) Steps: **1** Yield: **62%**



Suppliers (4)

Suppliers (8)

31-491-CAS-15714807 Steps: **1** Yield: **62%**

1.1 **Reagents:** Oxygen
Catalysts: Cuprous iodide
Solvents: Acetic acid, Dimethyl sulfoxide; 24 h, 130 °C

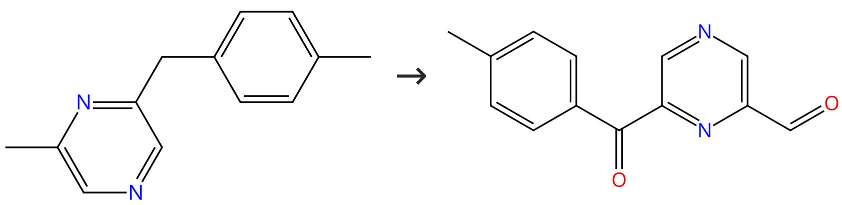
Experimental Protocols

Base metal-catalyzed benzylic oxidation of (aryl)(heteroaryl) methanes with molecular oxygen

By: Sterckx, Hans; et al

Beilstein Journal of Organic Chemistry (2016), 12, 144-153.

Scheme 11 (1 Reaction) Steps: **1** Yield: **61%**



31-491-CAS-15714771 Steps: **1** Yield: **61%**

1.1 **Reagents:** Oxygen
Catalysts: Cuprous iodide
Solvents: Acetic acid, Dimethyl sulfoxide; 24 h, 1 atm, 130 °C

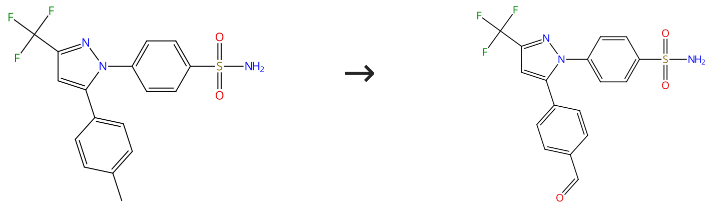
Experimental Protocols

Base metal-catalyzed benzylic oxidation of (aryl)(heteroaryl) methanes with molecular oxygen

By: Sterckx, Hans; et al

Beilstein Journal of Organic Chemistry (2016), 12, 144-153.

Scheme 12 (1 Reaction) Steps: **1** Yield: **41%**



Suppliers (120) Suppliers (8)

31-614-CAS-33763341 Steps: **1** Yield: **41%**

1.1 **Reagents:** Pyridine, Borate(1-), tetrafluoro-, hydrogen, compd. with pyridine (1:1:1)
Solvents: Acetonitrile, Dichloromethane; 50 min, rt

1.2 **Reagents:** Benzophenone, 4-(Dimethylamino)pyridine
Catalysts: *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: Acetone; 24 h, 22 - 27 °C

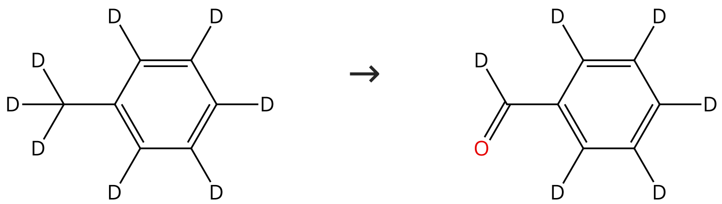
Experimental Protocols

Electrochemical PINOylation of Methylarenes: Improving the Scope and Utility of Benzylic Oxidation through Mediated Electrolysis

By: Hoque, Asmaul Md; et al

Journal of the American Chemical Society (2022), 144(33), 15295-15302.

Scheme 13 (1 Reaction) Steps: **1**



Suppliers (120) Suppliers (20)

31-614-CAS-35375748 Steps: **1**

1.1 **Reagents:** Oxygen
Catalysts: Methanesulfonic acid
Solvents: Acetonitrile; 1.5 h, 20 °C

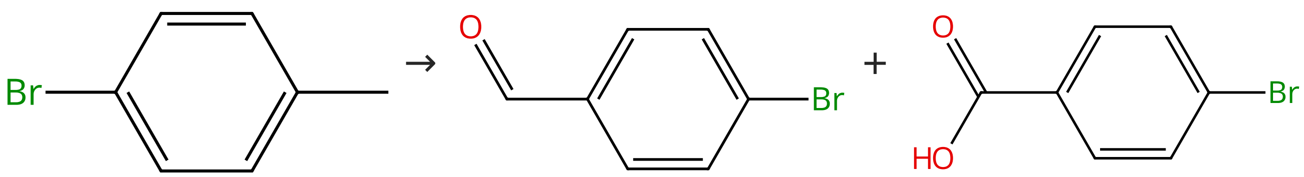
Experimental Protocols

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Scheme 14 (1 Reaction) Steps: **1** Yield: **41%**



Suppliers (67) Suppliers (90) Suppliers (98)

31-488-CAS-9304506 Steps: **1** Yield: **41%**

1.1 **Catalysts:** Rhodium, tetrakis[μ-[*N*-(3,4-dihydro-2*H*-pyrrol-5-yl-κ*N*)methanesulfonamidato-κ*N*]]di-, (*Rh*-*Rh*), stereoisomer
Solvents: Water; rt

1.2 **Reagents:** *tert*-Butyl hydroperoxide
Solvents: Water; 20 h, rt

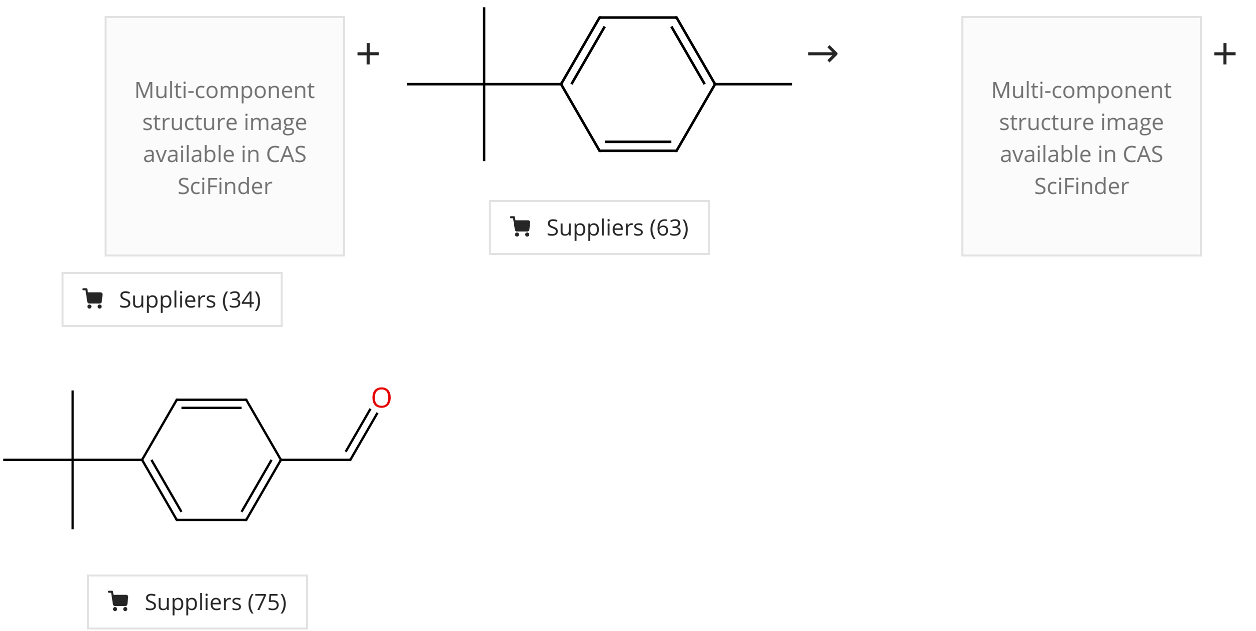
Experimental Protocols

Dirhodium(II) Complexes of 2-(Sulfonylimino)pyrrolidine: Synthesis and Application in Catalytic Benzylic Oxidation

By: Wusiman, Abudureheman; et al

European Journal of Organic Chemistry (2012), 2012(16), 3088-3092, S3088/1-S3088/45.

Scheme 15 (1 Reaction) Steps: **1** Yield: **7%**



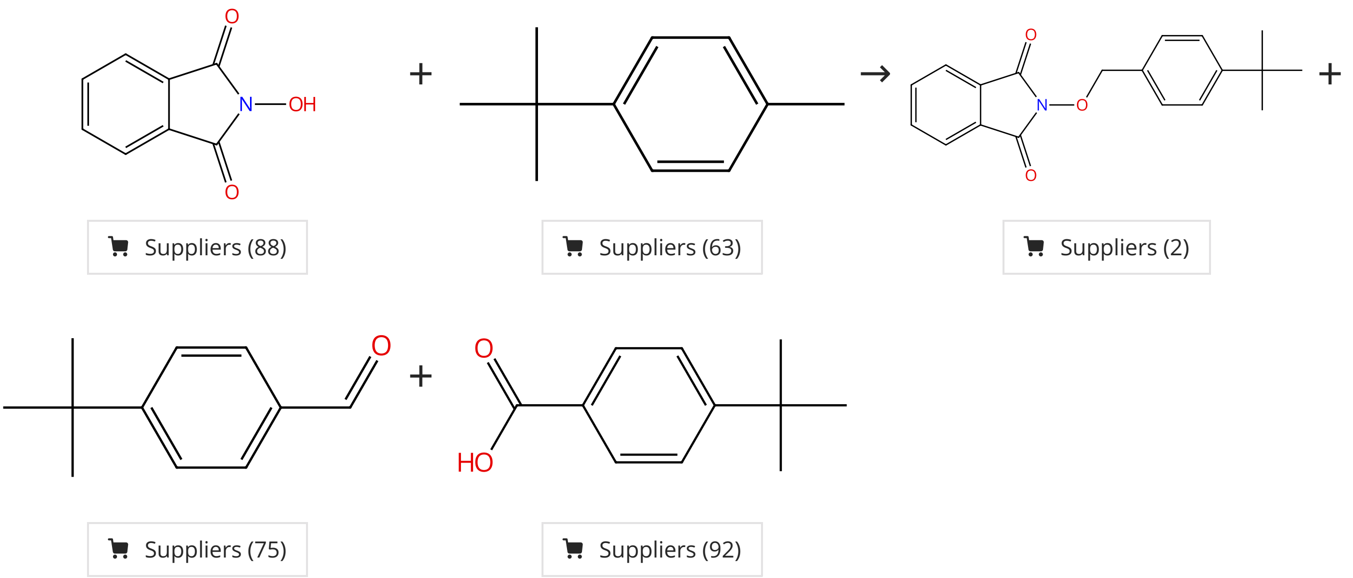
31-614-CAS-33763338	Steps: **1** Yield: **7%**
1.1	**Reagents:** Pyridine **Solvents:** Acetonitrile, Dichloromethane; 5.4 h, rt
Experimental Protocols	

Electrochemical PINOylation of Methylarenes: Improving the Scope and Utility of Benzylic Oxidation through Mediated Electrolysis

By: Hoque, Asmaul Md; et al

Journal of the American Chemical Society (2022), 144(33), 15295-15302.

Scheme 16 (1 Reaction) Steps: **1** Yield: **9%**



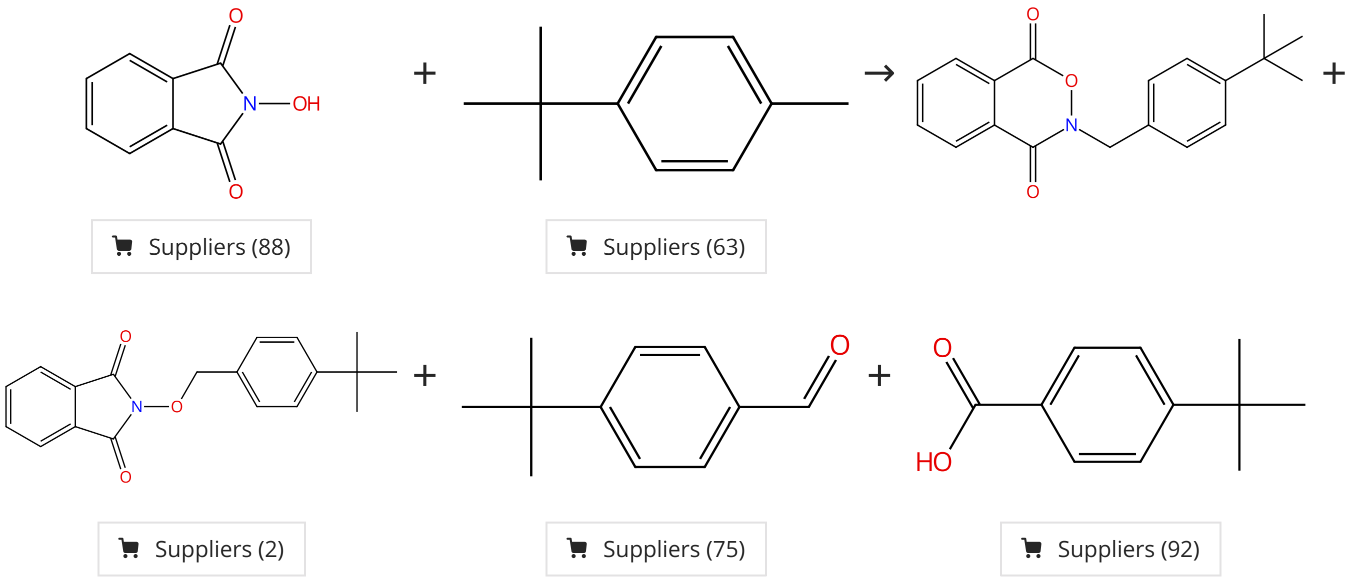
31-614-CAS-33763331	Steps: **1** Yield: **9%**
1.1	**Reagents:** Pyridine, Borate(1-), tetrafluoro-, hydrogen, compd. with pyridine (1:1:1), Oxygen **Solvents:** Acetonitrile, Dichloromethane; 40 - 60 min, rt
Experimental Protocols	

Electrochemical PINOylation of Methylarenes: Improving the Scope and Utility of Benzylic Oxidation through Mediated Electrolysis

By: Hoque, Asmaul Md; et al

Journal of the American Chemical Society (2022), 144(33), 15295-15302.

Scheme 17 (1 Reaction) Steps: **1** Yield: **10%**



31-614-CAS-33763334 Steps: **1** Yield: **10%**

1.1 **Reagents:** Pyridine, Borate(1-), tetrafluoro-, hydrogen, compd. with pyridine (1:1:1), Oxygen
Solvents: Acetonitrile, Dichloromethane; 40 - 60 min, rt

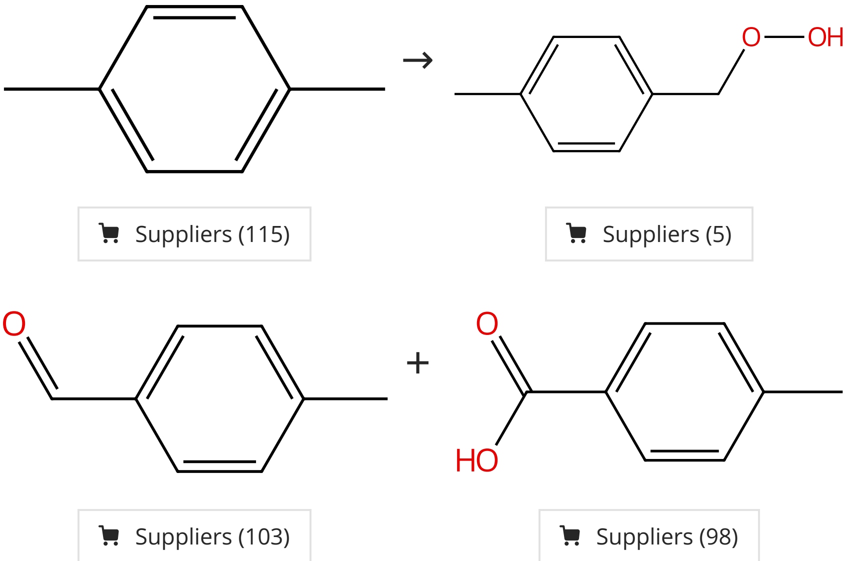
Experimental Protocols

Electrochemical PINOylation of Methylarenes: Improving the Scope and Utility of Benzylic Oxidation through Mediated Electrolysis

By: Hoque, Asmaul Md; et al

Journal of the American Chemical Society (2022), 144(33), 15295-15302.

Scheme 18 (2 Reactions) Steps: **1**



31-207-CAS-15199607 Steps: **1**

1.1 **Catalysts:** Graphite

Benzylic oxidation and photooxidation by air in the presence of graphite and cyclohexene

By: Sereda, Grigoriy; et al

Tetrahedron Letters (2007), 48(19), 3417-3421.

31-207-CAS-13070619 Steps: **1**

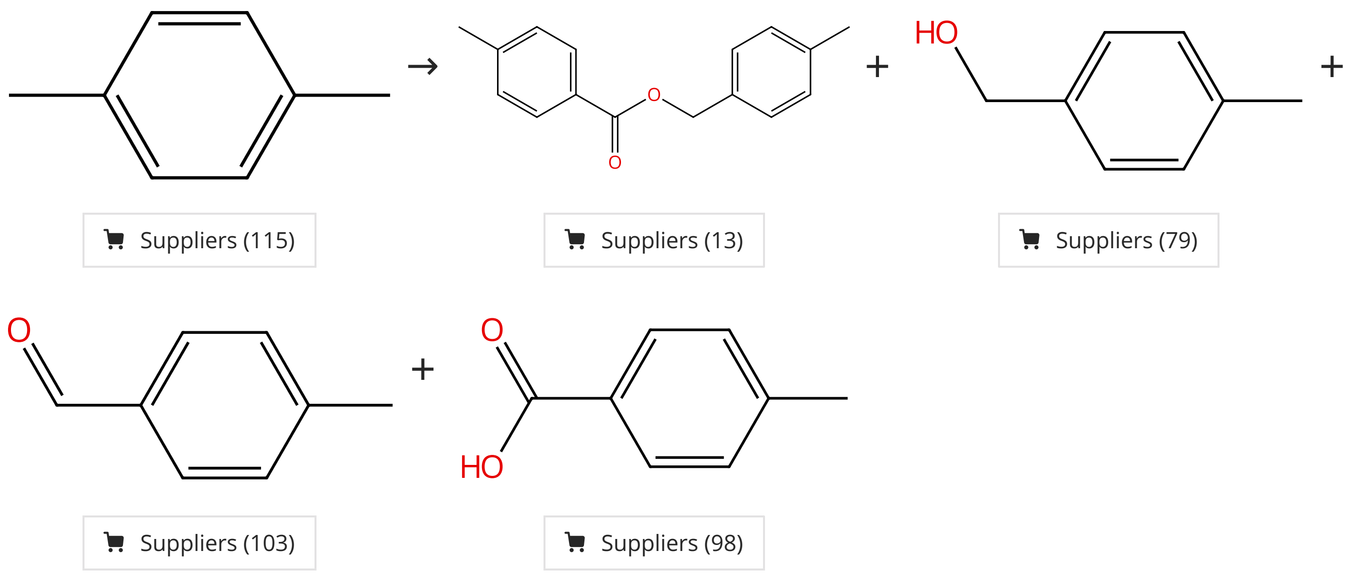
1.1 **Reagents:** Cyclohexene, Oxygen

Benzylic oxidation and photooxidation by air in the presence of graphite and cyclohexene

By: Sereda, Grigoriy; et al

Tetrahedron Letters (2007), 48(19), 3417-3421.

Scheme 19 (1 Reaction) Steps: **1**



31-488-CAS-10933722 Steps: **1**

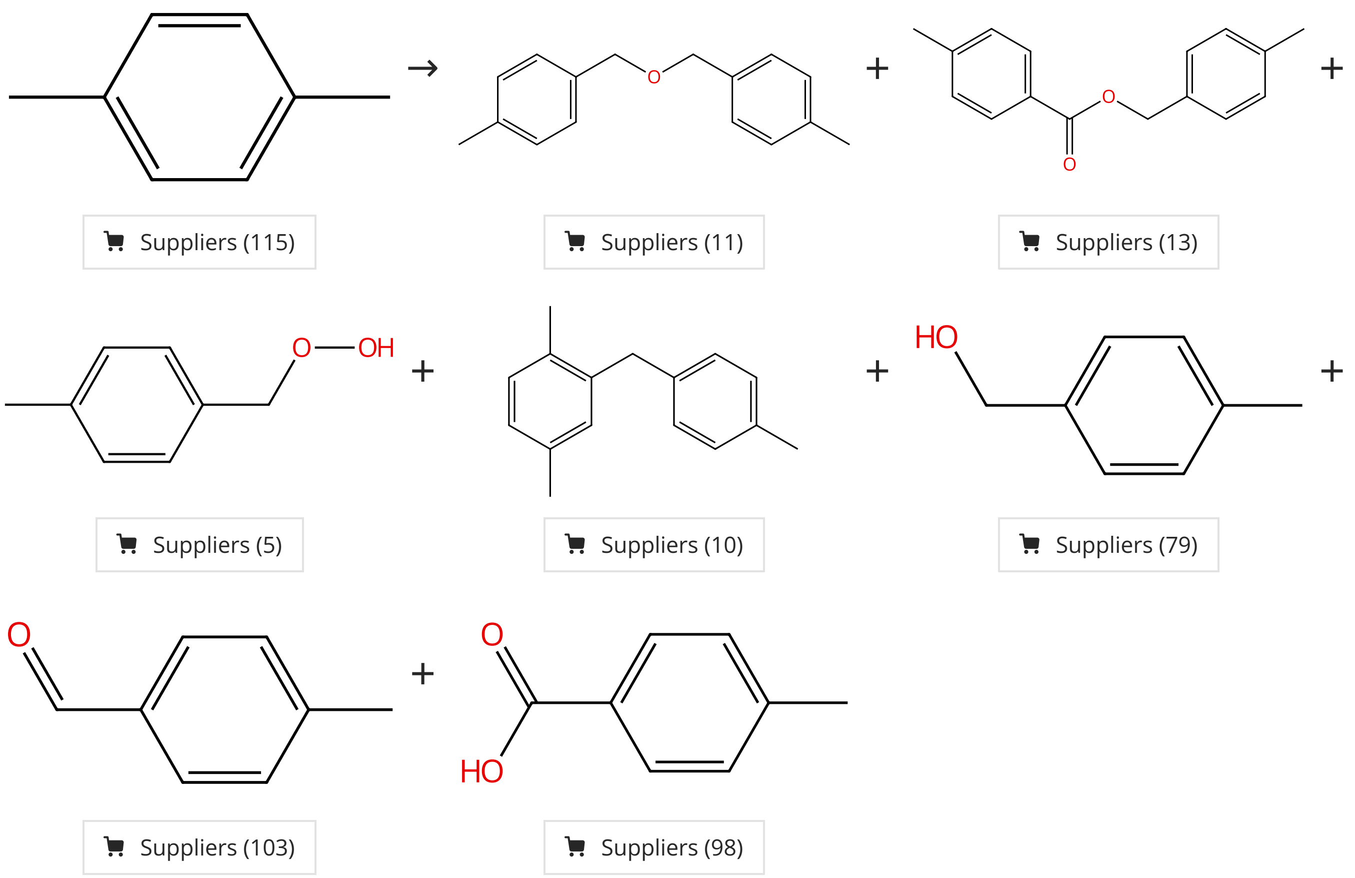
1.1 **Reagents:** Cyclohexene, Oxygen
Catalysts: Graphite

Benzylic oxidation and photooxidation by air in the presence of graphite and cyclohexene

By: Sereda, Grigoriy; et al

Tetrahedron Letters (2007), 48(19), 3417-3421.

Scheme 20 (1 Reaction) Steps: **1**



31-207-CAS-10834198 Steps: **1**

1.1 **Reagents:** Oxygen

Photoactivated and photopassivated benzylic oxidation catalyzed by pristine and oxidized carbons

By: Sereda, Grigoriy; et al

Catalysis Communications (2011), 12(7), 669-672.